Composition profile with upstream $a = a_N$

$a(x)$

Velocity profile $u(x)$

$u(x)$ [m/s]

Baryon density profile

$n_B(x)$ [fm^{-3}]

μ_K branch split

$\mu_K(x)$ [MeV]

μ_B profile

$\mu_B(x)$ [MeV]

x [nm]

trial 3: Kaon rich, jB=0.485, aQ=0.177
trial 10: Kaon poor, jB=0.541, aQ=0.196
trial 13: Kaon rich, jB=0.739, aQ=0.257
trial 14: Kaon poor, jB=0.738, aQ=0.257
$x < 0$: upstream constant state
$x > 0$: quark profile